

Отчет по домашнему заданию

Вариант 2: сбор метрик ClickHouse через Prometheus endpoint

Цель работы: настроить экспорт метрик ClickHouse в формате Prometheus и подтвердить получение любой метрики скриншотом результата сбора.

1. Краткая информация

Сервер	compute-vm-2-4-30-ssd-1776951461260
Пользователь	yc-user
Endpoint метрик	http://localhost:9363/metrics
Проверенная метрика	ClickHouseProfileEvents_Query

2. Конфигурация ClickHouse

Для отдачи метрик был создан конфигурационный файл:

```
/etc/clickhouse-server/config.d/prometheus.xml
```

Содержимое файла:

```
<clickhouse>
  <prometheus>
    <endpoint>/metrics</endpoint>
    <port>9363</port>
    <metrics>true</metrics>
    <events>true</events>
    <asynchronous_metrics>true</asynchronous_metrics>
    <status_info>true</status_info>
  </prometheus>
</clickhouse>
```

3. Команды настройки и проверки

После внесения конфигурации был выполнен перезапуск сервиса ClickHouse:

```
sudo systemctl restart clickhouse-server
```

Проверка доступности endpoint с метриками:

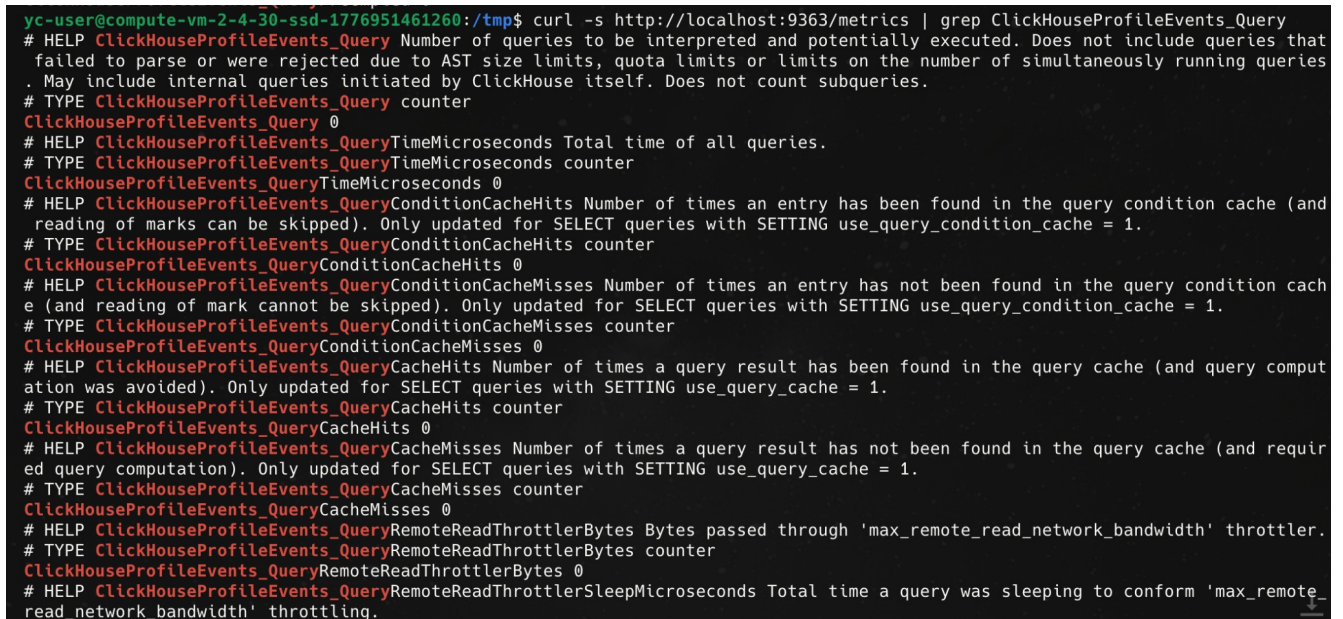
```
curl -s http://localhost:9363/metrics | grep ClickHouseProfileEvents_Query
```

4. Результат сбора метрики

В результате проверки endpoint вернул метрики в Prometheus-формате. В выводе присутствует метрика ClickHouseProfileEvents_Query:

```
# HELP ClickHouseProfileEvents_Query Number of queries to be interpreted and potentially executed.
# TYPE ClickHouseProfileEvents_Query counter
ClickHouseProfileEvents_Query 0
```

Скриншот результата проверки:



```
yc-user@compute-vm-2-4-30-ssd-1776951461260:/tmp$ curl -s http://localhost:9363/metrics | grep ClickHouseProfileEvents_Query
# HELP ClickHouseProfileEvents_Query Number of queries to be interpreted and potentially executed. Does not include queries that
failed to parse or were rejected due to AST size limits, quota limits or limits on the number of simultaneously running queries
. May include internal queries initiated by ClickHouse itself. Does not count subqueries.
# TYPE ClickHouseProfileEvents_Query counter
ClickHouseProfileEvents_Query 0
# HELP ClickHouseProfileEvents_QueryTimeMicroseconds Total time of all queries.
# TYPE ClickHouseProfileEvents_QueryTimeMicroseconds counter
ClickHouseProfileEvents_QueryTimeMicroseconds 0
# HELP ClickHouseProfileEvents_QueryConditionCacheHits Number of times an entry has been found in the query condition cache (and
reading of marks can be skipped). Only updated for SELECT queries with SETTING use_query_condition_cache = 1.
# TYPE ClickHouseProfileEvents_QueryConditionCacheHits counter
ClickHouseProfileEvents_QueryConditionCacheHits 0
# HELP ClickHouseProfileEvents_QueryConditionCacheMisses Number of times an entry has not been found in the query condition cach
e (and reading of mark cannot be skipped). Only updated for SELECT queries with SETTING use_query_condition_cache = 1.
# TYPE ClickHouseProfileEvents_QueryConditionCacheMisses counter
ClickHouseProfileEvents_QueryConditionCacheMisses 0
# HELP ClickHouseProfileEvents_QueryCacheHits Number of times a query result has been found in the query cache (and query comput
ation was avoided). Only updated for SELECT queries with SETTING use_query_cache = 1.
# TYPE ClickHouseProfileEvents_QueryCacheHits counter
ClickHouseProfileEvents_QueryCacheHits 0
# HELP ClickHouseProfileEvents_QueryCacheMisses Number of times a query result has not been found in the query cache (and requir
ed query computation). Only updated for SELECT queries with SETTING use_query_cache = 1.
# TYPE ClickHouseProfileEvents_QueryCacheMisses counter
ClickHouseProfileEvents_QueryCacheMisses 0
# HELP ClickHouseProfileEvents_QueryRemoteReadThrottlerBytes Bytes passed through 'max_remote_read_network_bandwidth' throttler.
# TYPE ClickHouseProfileEvents_QueryRemoteReadThrottlerBytes counter
ClickHouseProfileEvents_QueryRemoteReadThrottlerBytes 0
# HELP ClickHouseProfileEvents_QueryRemoteReadThrottlerSleepMicroseconds Total time a query was sleeping to conform 'max_remote
read_network_bandwidth' throttling.
```

5. Вывод

Экспорт метрик ClickHouse в формате Prometheus настроен. Endpoint `http://localhost:9363/metrics` доступен и возвращает метрики. Проверена метрика `ClickHouseProfileEvents_Query`, что подтверждает работоспособность сбора метрик со стороны ClickHouse.

Примечание: значение 0 у счетчика не является ошибкой. Оно означает, что на момент проверки счетчик запросов не увеличивался или был равен нулю.